



AI-Powered Student-Job Matching Fairness Audit & Model Performance Report

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Executive Summary

This report presents a comprehensive fairness audit and machine learning model performance analysis for PlaceMux, an AI-powered student-job matching platform. The analysis covers model accuracy, fairness metrics across demographic groups, and recommendations for deployment.

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2. Model Performance Metrics

The following metrics were computed on the held-out test set to evaluate model performance on unseen data. These metrics are crucial for understanding the reliability and effectiveness of the matching algorithm.

Metric	Value	Interpretation
Accuracy	87.0%	Proportion of correct predictions
Precision	85.0%	Accuracy of positive predictions
Recall	89.0%	Coverage of actual positives
F1 Score	87.0%	Harmonic mean of precision & recall
ROC-AUC	91.0%	Classification performance across thresholds
False Positive Rate	8.0%	Incorrect positive predictions

Interpretation: With an accuracy of 87.0% and F1 score of 87.0%, the model demonstrates reliable performance on unseen data. The precision of 85.0% indicates that when the model recommends a match, it is correct 85.0% of the time.

3. Fairness Audit Results

Fairness Metric	Score	Status
Overall Fairness	82.5/100	✓ EXCELLENT
Disparate Impact	85.0/100	✓ EXCELLENT
Predictive Parity	80.0/100	✓ EXCELLENT
Equalized Odds	82.0/100	✓ EXCELLENT

Overall Assessment: ✓ GOOD - Monitor in production

Fairness Concepts:

- **Disparate Impact:** Checks if recommendation rates differ significantly across groups (4/5 rule)
- **Predictive Parity:** Ensures positive prediction rate is consistent across demographics
- **Equalized Odds:** Verifies false positive and false negative rates are equal across groups

4. Demographic Analysis

GENDER

Group	Count	Accuracy	Recall	PPR
M	600	87.00%	88.00%	42.00%
F	400	87.00%	91.00%	44.00%

REGION

Group	Count	Accuracy	Recall	PPR
Metro	400	88.00%	90.00%	43.00%
Tier1	300	87.00%	88.00%	42.00%
Tier2	200	85.00%	87.00%	41.00%
Tier3	100	83.00%	85.00%	40.00%

5. Recommendations & Conclusions

Key Recommendations:

1. **Deployment Readiness:** With strong model performance and fairness metrics, the system is ready for pilot deployment in a controlled environment.
2. **Continuous Monitoring:** Implement real-time monitoring of predictions across demographic groups to detect any drift in fairness metrics post-deployment.
3. **Model Retraining:** Establish a quarterly retraining schedule with updated data to maintain model performance and fairness as the student and job populations evolve.
4. **Explainability:** Ensure all predictions provided to users include clear explanations of the matching criteria to build trust and allow for human review.
5. **Privacy & Consent:** Maintain DPDP compliance by ensuring explicit consent for data usage and easy data deletion mechanisms for all users.
6. **Bias Mitigation:** Continue monitoring for emerging biases. Consider collecting additional demographic data to enable intersectional analysis.

Conclusion:

PlaceMux demonstrates strong potential as an AI-powered recruitment tool that balances accuracy with fairness. The comprehensive fairness audit confirms that the system maintains equitable treatment across demographic groups while delivering reliable match predictions. With continued monitoring and the recommended improvements, PlaceMux can become a trusted platform for student-job matching.

6. Technical Specifications

Model Architecture:

- Algorithm: Random Forest Classifier (100 estimators)
- Feature Count: 17 engineered features
- Training Samples: ~800 (80% of 1000 total)
- Test Samples: ~200 (20% of 1000 total)

Features Used:

- Student: Years of experience, GPA, projects, internships, skill scores
- Job: Experience requirements, GPA requirement, salary range, skills
- Computed: Skill match ratios, compatibility scores

Data Preprocessing:

- Categorical encoding: Label Encoding for 5 categorical features
- Feature scaling: StandardScaler for model training
- Stratified train-test split to preserve class distribution

Fairness Assessment Framework:

- Demographic parity (disparate impact) analysis
- Predictive parity across groups
- Equalized odds evaluation
- False positive and false negative rate analysis

This report is auto-generated by PlaceMux Reporting System. For questions or clarifications, please contact the AI/ML Engineering team.